Molecular Cloning

This part documents all the operations involved in plasmid construction for our workflow.

This section includes:

1. Plasmid extraction
2. Agarose gel electrophoresis
3. Gel extraction
4. Transformation of competent cells
5. Plasmid construction by homologous recombination

Protocol – Molecular Cloning

Plasmid extraction

This protocol follows the manual of the TIANprep Mini Plasmid Kit (TIANGEN Biotech, Beijing). Before use: add the entire vial of RNase A supplied with the kit to buffer P1, mix well, and store at 2–8°C. Check whether equilibration buffer BL, buffer P2, and buffer P3 have gone turbid; if so, warm them in a 37°C water bath for 10 min and use them once they have cleared up again. Add anhydrous ethanol to wash buffer PW, in the volume given on the bottle label.

1. Column equilibration: add 500 µL of equilibration buffer BL to adsorption column CP3 (sitting in a collection tube), centrifuge at 12,000 rpm (~13,400 × g) for 1 min, discard the flow-through, and put the column back into the collection tube.
2. Transfer 1–5 mL of an overnight bacterial culture into a centrifuge tube and spin it in a benchtop centrifuge at 12,000 rpm (~13,400 × g) for 1 min; remove as much of the supernatant as you can. For larger culture volumes, spin in several rounds and pool the bacterial pellets into the same tube.
3. Add 250 µL of buffer P1 to the cell pellet (check first that RNase A has been added), and resuspend the pellet completely with a pipette or on a vortex.
4. Add 250 µL of buffer P2 to the centrifuge tube, and gently invert the tube 6–8 times to fully lyse the cells. Note: keep the mixing gentle — never shake it hard, or the genomic DNA will shear and contaminate the plasmid prep. At this point the lysate should look clear and viscous. Don't let the lysis run past 5 min, or the plasmid itself gets damaged. If the suspension doesn't clear up, there was too much bacteria — cut down the culture volume next time.
5. Add 350 µL of buffer P3 and immediately, gently invert the tube 6–8 times to mix; a white flocculent precipitate should appear. Centrifuge at 12,000 rpm (~13,400 × g) for 10 min.
6. Draw up the supernatant from the previous step and load it onto adsorption column CP3 (in a collection tube), taking care not to touch the precipitate. Centrifuge at 12,000 rpm (~13,400 × g) for 30–60 s, discard the flow-through, and return the column to the collection tube.
7. Optional step: add 500 µL of deproteinization buffer PD to column CP3, centrifuge at 12,000 rpm (~13,400 × g) for 60 s, pour off the waste, and put the column back into the collection tube.
8. Add 600 µL of wash buffer PW to column CP3 (check first that anhydrous ethanol has been added), centrifuge at 12,000 rpm (~13,400 × g) for 30–60 s, discard the waste, and put the column back into the collection tube.
9. Repeat step 8.

10. Put column CP3 back into the collection tube and centrifuge at 12,000 rpm (~13,400 × g) for 2 min to drive off the residual wash buffer. Note: ethanol left in the wash buffer will interfere with downstream enzymatic reactions such as restriction digestion and PCR. To keep residual ethanol from affecting the downstream work, open the cap and let the column sit at room temperature for a few minutes so that the adsorption membrane dries completely.
11. Place column CP3 into a clean centrifuge tube, add 50–100 μL of elution buffer to the center of the adsorption membrane, leave it at room temperature for 2 min, then centrifuge at 12,000 rpm (~13,400 × g) for 2 min and collect the plasmid solution in the tube. Note: never elute with less than 50 μL — small volumes drag down the recovery. The pH of the eluent has a strong effect on elution efficiency: if the plasmid is headed for sequencing, elute it in double-distilled water held at pH 7.0–8.5; below pH 7.0 the yield drops. Keep the DNA product at −20°C so it doesn't degrade. For a higher plasmid recovery, reload the collected eluate onto the column, leave it for 2 min at room temperature, and spin at 12,000 rpm (~13,400 × g) for 2 min to collect it again.

Agarose gel electrophoresis

1. Gel casting

1.1 Set the agarose concentration from the size of the DNA fragments to be separated, and calculate the agarose mass from the gel volume and concentration.

1.2 Weigh out the agarose and add it to the calculated volume of 1× TAE buffer.

1.3 Heat the solution in a microwave until it boils and turns completely transparent, with no visible suspended particles.

1.4 Once the gel solution has cooled to about 50–60°C, add GelRed nucleic acid stain to a final concentration of 1% and mix well.

1.5 Pour the gel slowly into the casting tray and insert the comb in position. Keep the tray level so that the gel spreads evenly, with no bubbles trapped.

2. Sample preparation

Mix each DNA sample with loading buffer at the proper ratio (the exact ratio depends on the dilution of the loading buffer in use).

3. Loading

3.1 Place the set gel into the electrophoresis tank and pour in running buffer until the gel is fully submerged.

3.2 Load the samples and the DNA molecular weight marker into the wells with a micropipette; switch tips after every sample to avoid cross-contamination. Adjust the load volume to the sample concentration, and stay within the well capacity.

4. Running

4.1 Close the lid of the tank and double-check the electrodes: the negative electrode (black) connects on the well side, and the positive electrode (red) on the far end.

4.2 Set the voltage on the power supply — usually 90–150 V — and adjust the run time to the fragment size and the resolution needed, typically 30 min to 2 h. Stop the run once the bands have migrated to about 1–2 cm from the gel front.

5. Reading the gel

Put the gel under a UV imaging system or another suitable detector (DNA bound to the nucleic acid stain fluoresces under UV excitation, which is how band positions and fragment sizes are read), and record the result with a gel documentation system.

Gel extraction

This protocol follows the manual of the E.Z.N.A.® Gel Extraction Kit (spin-column type) from Omega Bio-Tek.

Setup before the experiment: dilute SPW buffer with anhydrous ethanol and keep it at room temperature.

1. Separate the DNA fragments by agarose gel electrophoresis (ethidium bromide staining works); any grade of agarose is acceptable. Freshly prepared TAE or TBE running buffer is strongly recommended — reusing running buffer raises its pH and cuts into DNA recovery.
2. Once the bands are fully separated, cut out the target band with a clean, sharp scalpel, and trim away as much extra agarose as possible to keep the gel slice small.
3. Transfer the gel slice into a 1.5 mL centrifuge tube, weigh it, and note down the mass.
4. Add XP2 binding buffer in a volume equal to that of the gel slice. Note: the XP2 volume goes as 1 mL per 1 g of gel — for example, a 0.3 g slice takes 0.3 mL.
5. Incubate at 50–60°C for 7 min, until the slice has fully melted; vortex the tube every 2–3 min during the incubation.
6. Seat a HiBind® DNA mini column in a 2 mL collection tube.
7. Apply the DNA–agarose solution from step 5 onto the HiBind® DNA mini column, with no more than 700 µL per load; centrifuge at 10,000 × g for 1 min at room temperature, discard the filtrate, and reuse the collection tube.
8. Repeat step 7 until the whole sample has been transferred onto the column.
9. Discard the filtrate, put the HiBind® DNA mini column back into the 2 mL collection tube, add 300 µL of XP2 binding buffer, centrifuge at maximum speed (≥13,000 × g) for 1 min at room temperature, and discard the filtrate.
10. Return the HiBind® DNA mini column to the 2 mL collection tube, add 700 µL of SPW buffer (already diluted with ethanol as instructed), centrifuge at 10,000 × g for 1 min at

room temperature, and discard the filtrate. Optional: if the downstream experiment is sensitive to salt carryover, repeat this wash once more with SPW buffer.

11. Put the HiBind® DNA mini column back into the 2 mL collection tube and centrifuge at ≥13,000 × g for 2 min to dry the column matrix completely.
12. Transfer the HiBind® DNA mini column to a clean 1.5 mL centrifuge tube and add 15–30 μL of elution buffer or deionized water directly onto the center of the matrix; leave it at room temperature for 2 min, then centrifuge at 13,000 × g for 1 min to elute the DNA. Note: the first elution typically recovers 70–80% of the DNA bound to the column, and a second elution can be done if needed. Eluting a second time with fresh buffer raises the total yield but dilutes the DNA; re-running the first eluate through the column instead concentrates the DNA without adding volume.

Transformation of competent cells

1. Thaw the Trelief 5α competent cells on ice; add 10 μL of the homologous recombination product to each 100 μL aliquot of competent cells, and keep the mix on ice for 30 min.
2. Heat-shock at 42°C for 60 s, then put the tubes back on ice for 2 min. Add 700 μL of antibiotic-free LB medium and let the cells recover at 37°C, 200 rpm for 60 min.
3. Centrifuge at 5,000 rpm for 2 min, leave about 200 μL of the supernatant behind to resuspend the cells, and plate the suspension onto LB plates containing kanamycin.

Plasmid construction by homologous recombination

1. Preparing the linearized vector (inverse PCR)

1.1 Pick a suitable cloning site to linearize the vector. Sites are preferred where the 20 bp flanking the cloning site are free of repeated sequences and the GC content sits at 40–60%.

1.2 Primer design: the primers bind at the two ends of the known sequence and extend in the direction opposite to a normal PCR — the forward primer binds downstream of the known sequence, the reverse primer upstream, and both extend toward the unknown region. Primer sequences are homologous to the template strand, 18–25 bp long, Tm 55–65°C, and designed to steer clear of secondary structures.

1.3 Set up the reaction system and the cycling program per the KOD One™ PCR Master Mix-Blue manual.

Reaction system:

Component	**Volume**
Sterile water	21 − X μL
KOD One™ PCR Master Mix-Blue	25 μL
Forward primer F (10 μM)	2 μL

Reverse primer R (10 μM)	2 μL
Template DNA	X μL (50 ng total)

Cycling program:

Stage	**Temperature**	**Time**	**Cycles**
Initial denaturation	98°C	5 min	—
Denaturation	98°C	10 s	34
Annealing	X°C	5 s	34
Extension	68°C	Y min	34
Final extension	68°C	5 min	—
Hold	4°C	∞	—

Notes: X = primer Tm − 5°C; Y is set from the length of the target fragment — 5 s for fragments under 1 kb, 5 s/kb for fragments of 1–10 kb, and 10 s/kb above 10 kb.

2. Getting the insert fragment

Take the 15–20 bp at the ends of the linearized vector as the homology-arm sequences, and append them to the 5′ ends of the gene-specific forward and reverse primers respectively. Amplifying with this primer pair yields an insert fragment carrying homology sequences at both ends. PCR system and program as above.

3. Agarose gel electrophoresis

Run the PCR products from the steps above on a gel to check whether the amplification worked; see the agarose gel electrophoresis section for details.

4. Gel extraction

Recover the PCR products from the gel by gel extraction; see part 3 (gel extraction) for details.

5. Homologous recombination

Mix the linearized vector and the insert fragment in the following system:

Component	**Amount**
Linearized vector	0.04 × total vector bp ÷ vector concentration (ng/μL), in μL
Insert fragment	0.04 × total insert bp ÷ insert concentration (ng/μL), in μL
ClonExpress recombinase mix (2×)	5 μL
Double-distilled water	top up to 10 μL

Note: if the linearized vector is too concentrated, dilute it before use. Catalyzed by the 2× ClonExpress recombinase mix, a 30 min incubation at 50°C completes the recombination, circularizing the two linear DNA molecules in vitro.

6. Transformation of competent cells

7. Verification of positive clones

7.1 Culture expansion

Pick single colonies that are distinct, well separated, and not overlapping, and inoculate them at 1:1000 into LB liquid medium with the corresponding antibiotic. Grow for 5–8 h until the culture turns turbid.

7.2 Colony PCR

Design primers for colony PCR to make a first check of whether the target fragment made it into the plasmid; specific primers can also be used to confirm which strain was picked (for example, donor-specific primers to check for *Bacillus subtilis*). Set up the reaction system and the program per the 2× Rapid Taq Master Mix manual.

Reaction system:

Component	**Volume**
Sterile water	7 μL
2× Rapid Taq Master Mix	10 μL
Forward primer F (10 μM)	1 μL
Reverse primer R (10 μM)	1 μL
Cell suspension template	1 μL

Cycling program:

Stage	**Temperature**	**Time**	**Cycles**
Initial denaturation	95°C	3 min	—
Denaturation	95°C	15 s	34
Annealing	X°C	15 s	34
Extension	72°C	Y min	34
Final extension	72°C	5 min	—
Hold	4°C	—	—

Notes: X = primer Tm − 5°C; Y follows the length of the target fragment, at 15 s/kb.

7.3 Sequencing

Send the samples, together with the sequencing primers, to a sequencing company to verify that the plasmid sequence carries the target gene.

Osmotic Shock Transformation of D. salina

To get our target genes expressed in Dunaliella salina, we used an osmotic shock transformation method to deliver the assembled plasmids into the cells.

This section includes:

1. Cell culture and preconditioning
2. Screening and identification of transformants

Protocol – Osmotic Shock Transformation of D. salina

Cell culture and preconditioning

D. salina FACHB-435 was grown statically in modified DM medium (with 1.5 M NaCl) at 26°C, under 50 μmol photons m^{-2} s^{-1} of light and a 16 h daily photoperiod. The cells were harvested once they reached logarithmic phase (about 10^5–10^6 cells/mL), at which point the OD680 reads roughly 1.0.

1. Take the log-phase culture and pellet the cells in a centrifuge.
2. Pour off the supernatant, resuspend the pellet in fresh PKS/DM medium, and wash the cells 3 times to strip off residual metabolites from the old medium.
3. Resuspend the washed cells in fresh medium and adjust the final cell density to 10^6 cells/mL.
4. Transfer this suspension into fresh medium containing 1 M NaCl and keep culturing for another 8 h. This step "pre-adapts" the cells to the osmotic shock that follows.
5. Take 1 mL of the preconditioned suspension, spin it down, and discard the supernatant.
6. Immediately resuspend the pellet in fresh DM medium containing 0.1 M NaCl (bringing the volume back to 1 mL).
7. Add the following to this 1 mL suspension, in order and precisely:
    a) Plasmid DNA: 5.46 μg of the vector carrying the target gene
    b) Surfactant: 15 μL of 0.1% Triton X-100 stock
8. Vortex the mixture vigorously for about 90 s.
9. As soon as the vortexing is done, transfer the mixture into fresh liquid medium.
10. Let the cells recover statically for 48 h at 26°C under a 16 h daily photoperiod. By then, over 90% of the surviving transformed cells can return to normal growth.

Screening and identification of transformants

Fluorescence microscopy: cells from the recovery culture were collected and fixed with 4% paraformaldehyde for 30 min, washed twice by resuspension in PBS, and then examined under a fluorescence microscope. Because the vector carries EGFP, successfully transformed cells show green fluorescence.

Molecular-level identification (PCR): genomic DNA was extracted from the transformed cells and amplified with gene-specific primers. A band of the expected size confirms that the exogenous gene has been stably integrated into the cell genome.

Selection steps:

1. DM solid medium with a salt concentration of 0.5 M and 2% agar
2. Antibiotic concentration screen: take the same amount of D. salina as used after the osmotic transformation culture, collect the algal pellet by low-speed centrifugation, and plate it on D. salina plates with different antibiotics to pin down the minimum lethal concentration
3. Plate the post-transformation algae onto plates with the corresponding antibiotic; after they have grown for a while, pick single colonies and scale them up — these are the transformants

4. Extract the genomic DNA again and verify the transformants by qPCR

Plant Genomic DNA Extraction

To check whether the target gene had actually integrated into the chromosomal genome of D. salina, we extracted genomic DNA from the algae.

This section includes:

1. DNA extraction
2. DNA quantification and purity check

Protocol –Plant Genomic DNA Extraction

Genomic DNA extraction

We used the Plant Genomic DNA Kit from TIANGEN.

Before starting, add anhydrous ethanol to buffer GD and wash buffer PW — the volumes are specified on the bottle labels.

1. Take about 100 mg of fresh D. salina algal pellet (or ~30 mg of dried tissue) and grind it thoroughly in liquid nitrogen.
2. Quickly transfer the ground powder into a centrifuge tube pre-loaded with 700 μL of buffer GP1 pre-warmed to 65°C (β-mercaptoethanol was added to the pre-warmed GP1 beforehand, to a final concentration of 0.1%). Mix right away by inverting, then incubate the tube in a 65°C water bath for 20 min, inverting the tube a few times along the way to keep the sample mixed.
3. Add 700 μL of chloroform, mix thoroughly, and centrifuge at 12,000 rpm (~13,400 × g) for 5 min. Note: for tissues rich in polyphenols or starch, do an equal-volume phenol:chloroform (1:1) extraction before this step.
4. Carefully transfer the upper aqueous phase from the previous step into a fresh centrifuge tube, add 700 μL of buffer GP2, and mix well.
5. Load the mixture onto adsorption column CB3, centrifuge at 12,000 rpm (~13,400 × g) for 30 s, and discard the flow-through. (The column holds about 700 μL, so it can be loaded and spun in batches.)
6. Add 500 μL of buffer GD to column CB3 (double-check that anhydrous ethanol has been added), centrifuge at 12,000 rpm (~13,400 × g) for 30 s, pour off the waste, and put the column back into the collection tube.
7. Add 600 μL of wash buffer PW to column CB3 (double-check that anhydrous ethanol has been added), centrifuge at 12,000 rpm (~13,400 × g) for 30 s, pour off the waste, and put the column back into the collection tube.
8. Repeat step 7.
9. Put column CB3 back into the collection tube and centrifuge at 12,000 rpm (~13,400 × g) for 2 min; pour off the waste. Then leave the column open at room temperature for a few minutes so that any residual wash buffer in the adsorption material dries off completely. Note: the point of this step is to remove the residual wash buffer from the column — leftover ethanol will interfere with downstream enzyme reactions (restriction digestion, PCR, and the like).
10. Transfer column CB3 into a clean centrifuge tube, drip 50–200 μL of elution buffer TE onto the center of the adsorption membrane, leave it at room temperature for 2–5 min, then centrifuge at 12,000 rpm (~13,400 × g) for 2 min and collect the solution in the tube. Note: don’t elute with less than 50 μL — small volumes hurt the recovery. The pH of the eluent matters a great deal: if ddH2O is used as the eluent, keep its pH between 7.0 and 8.5, since anything below 7.0 lowers

the elution efficiency. Store the DNA product at −20°C to keep it from degrading. For a higher genomic DNA yield, the collected solution can be reloaded onto column CB3, left for 2 min at room temperature, and spun again at 12,000 rpm (~13,400 × g) for 2 min.

DNA quantification and purity check

How large the recovered genomic DNA fragments come out depends on how long the sample was stored and how much shear it went through during handling. The recovered DNA can be checked on an agarose gel, with its concentration and purity read off a UV spectrophotometer.

DNA should give a strong absorption peak at OD260; an OD260 of 1 corresponds to roughly 50 μg/mL of double-stranded DNA, or 40 μg/mL of single-stranded DNA.

The OD260/OD280 ratio should sit between 1.7 and 1.9. If the DNA was eluted in ddH2O rather than elution buffer, the ratio tends to run low, because pH and ions shift the absorbance values — that doesn't by itself mean the DNA is impure.

Quantitative Real-Time PCR (qPCR)

To measure the copy number of the target gene in the D. salina genome, we ran qPCR.

This section includes:

8. Genomic DNA elimination reaction
9. Reverse transcription reaction
10. Subsequent qPCR

Protocol –Quantitative Real-Time PCR (qPCR)

cDNA was synthesized from total RNA using PrimeScript™ FAST RT reagent Kit with gDNA Eraser (RR092A, Takara) following manufacturer's instructions.

Genomic DNA Elimination Reaction

Component	Volume
8× gDNA Eraser Premix	2 μL
Genomic RNA template	1 μL
RNase Free H_2O	13 μL
Total volume	16 μL

Set up the reactions on ice and out of the light — pure water and primers go in first, with the PCR thermal profile: 42 ℃ for 2 min (alternatively 5 min at room temperature; 42 ℃ is recommended). Place on ice immediately after reaction.

Reverse Transcription Reaction

Add 4 μL of 5× RT Premix directly into the 16 μL gDNA-eliminated product from the previous step. Gently mix by pipetting and spin down briefly.

Reagent	Volume
gDNA elimination product (16 μL from last step)	16 μL
5× RT Premix	4 μL
Total volume	20 μL

RT thermal profile:

- 37 ℃, 10 min (fast reverse transcription, characteristic of this kit)
- 85 ℃, 5 sec (heat inactivation of RT enzyme to terminate reaction)
- Hold at 4 ℃

The product is cDNA. It can be used directly for qPCR or diluted with EASY Dilution II before loading. Store short-term at −20 ℃, aliquot and store long-term at −80 ℃; avoid repeated freeze-thaw cycles.

Subsequent qPCR

Prepare the reaction mix on ice. Set 3 technical replicates per sample and NTC (no-template control).

Reagent	Volume
TB Green Premix Ex Taq II (2×)	10 μL
Forward Primer (10 μM)	0.8 μL
Reverse Primer (10 μM)	0.8 μL
cDNA template	2μL
RNase Free H_2O	6.4 μL

Total volume	20 μL

qPCR amplification program (compatible with QuantStudio and other ABI instruments)

1. Initial denaturation: 95 ℃, 30 s
2. Amplification (40 cycles): 95 ℃ for 5 s; 60 ℃ for 30 s (combined annealing & extension; adjust temperature according to primer Tm)
3. Melt curve analysis: 95 ℃ for 15 s → 60 ℃ for 1 min → ramp slowly to 95 ℃ with fluorescence acquisition to verify amplification specificity

Data processing

1. Export the raw Ct values, melt curves, and amplification curves once the run is over.
2. Look over the melt curves: a single, sharp, symmetric peak means specific amplification; data with extra peaks or primer dimers don't count as valid.
3. Pool the Ct values from each group, work out the means and standard deviations, and trim outliers.
4. Use an endogenous D. salina housekeeping gene as the internal reference to correct for DNA loading and extraction variation across samples.
5. Convert to relative genome copy numbers of the exogenous target gene with the $2^{-\Delta\Delta Ct}$ method.
6. Compare the signal between wild-type and transgenic lines to rank them by exogenous gene copy number, and pick out the stable, high-expression engineered strains.